We thank the editor for handling our submission, and we are grateful to the reviewers for their careful reading of the manuscript and their constructive feedback. Their comments have strengthened the paper and helped us improve both its clarity and overall presentation. In this document, we provide detailed responses to each comment.

Reviewer 1

> The reproducibility of results has become an important issue across fields. It remains an open question whether Axelrod's second tournament, which contributed significantly to research on direct reciprocity, can be reproduced. In this manuscript, the authors adapt the surviving Fortran implementations to modern compilers and integrate them into the Axelrod-Python framework. Despite the stochasticity of the game setting, the reproduced results are consistent with the original ones. The authors also provide a detailed analysis of the tournament results. In addition, they explore additional tournaments by introducing extra strategies and incorporating noise to test the performance.

Reply:

We sincerely thank the reviewer for their positive assessment of our work and for recognizing its contributions.

> I have the following comments and questions about the manuscript:
>
> 1. Do the original 63 submitted strategies include classical benchmark strategies such as AllC, AllD, or WSLS? If so, it would be better to report their rankings.

Reply:

The original set of 63 submitted strategies did not include classical benchmark strategies such as AllC, AllD, or WSLS. The only strategies that would now be considered classical were Tit For Tat and Random. Motivated by this comment, and the related point raised below, we carried out an additional analysis to examine what happens when such benchmark strategies are included.

Changes:

We have added a sentence clarifying that classical benchmark strategies were not included in the original tournament (in the "Reproducing Axelrod's Second Tournament" section, under "Running the tournament").

> 2. Regarding the Extra Invitations section, the win rate of TFT exhibits non-monotonic fluctuations as the number of additional strategies increases. Is this behavior primarily driven by the distributional properties of the 209 Axelrod-Python strategies?
>
> Moreover, this section seems to be motivated by computational exploration. Therefore, in addition to showing that TFT and k42r remain relatively robust when additional strategies are introduced, I suggest presenting results that incorporate classical strategies and examining how they affect the performance of TFT and k42r.

Reply:

The reviewer raises two related points. First, they ask about the non-monotonic behavior of the winning probability of *TFT* as the number of additional strategies increases. Second, they suggest incorporating classical benchmark strategies into the analysis in order to better understand how the performance of *TFT* and `k42r` changes in extended tournaments.

To this end, we first consider the effect of introducing several classical strategies into the original tournament. We find that, among the strategies we consider, only Anti-*TFT* is capable of dethroning *TFT*. This occurs because Anti-*TFT* is able to exploit *TFT*, lowering its payoff relative to other strategies. At the same time, `k42r` is able to exploit Anti-*TFT*, achieving payoffs higher than those obtained through mutual cooperation and ultimately winning the tournament.

We then extend this analysis by considering all possible combinations of these classical strategies. In this setting, we again observe a non-monotonic relationship between the number of added strategies and the winning probability of *TFT*. However, unlike in the large-scale tournament analysis, here we can explicitly identify the mechanism driving this behavior.

In particular, whenever Anti-*TFT* is included in the tournament, *TFT* ranks second while `k42r` ranks first. As a result, the winning probability of *TFT* is determined entirely by the number of subsets that contain Anti-*TFT*. For example, when two additional strategies are introduced, there are 36 possible combinations, of which 7 include Anti-*TFT*. These are precisely the tournaments in which *TFT* fails to win. The same combinatorial symmetry appears for subset sizes seven and two, six and three, and so forth, producing the approximately symmetric non-monotonic pattern observed in the manuscript.

We believe this analysis sheds additional light on the large-scale tournament results. In particular, it suggests that the observed non-monotonic behavior is not simply a property of the distribution of strategies, but can emerge from the combinatorial interaction structure induced by specific subsets of opponents. At the scale of the full Axelrod-Python tournament, however, the number of possible interaction structures becomes extremely large, making it difficult to fully characterize the precise mechanisms responsible for the observed fluctuations.

Changes:

We now begin the revisiting section by considering the case where classical benchmark strategies are included in the tournament. We first examine the effect of introducing these strategies individually and simultaneously (Table 1), and then extend the analysis to all possible combinations of the classical strategies for subset sizes ranging from 2 to 8 (Table 2).

We also include a new figure (Fig. 4) showing that, although the winning probability of *TFT* decreases for intermediate subset sizes, *TFT* remains consistently associated with higher average payoffs across tournaments. Interestingly, the reduction in *TFT*'s effectiveness coincides with the largest variation in tournament cooperation levels, suggesting that *TFT* struggles most in environments with highly variable interaction structures.

Finally, we now discuss the non-monotonic behavior of *TFT* in the Axelrod-Python tournament setting. While the classical strategy analysis allows us to identify a concrete combinatorial mechanism driving this behavior, a complete understanding of the large-scale tournament results remains difficult due to the combinatorial explosion of possible interaction structures and opponent subsets.

3. In a tournament, a strategy's performance is evaluated against a specific distribution

> of opponents. In Axelrod's second tournament, this distribution is defined by the other 62 submitted strategies; in the large-scale experiments, the Axelrod-Python strategy set is also included. As both distributions are subjective choices, is it possible to construct an "objective" distribution of opponents and obtain a more "objective" performance ranking accordingly?

Reply:

In practice, the set of participating strategies determines the types of interactions that can occur within the tournament. As demonstrated by our extra invitation experiments, modifying the tournament composition can substantially change the resulting rankings.

However, this also raises the question of what exactly we mean by a "distribution of opponents" and what it would mean for such a distribution to be "objective". One possible interpretation is that the opponent set induces a distribution over interaction types. For example, in the Supplementary Information, we plot the distributions of cooperation rates and payoffs across pairwise interactions for both the original tournament and the Axelrod-Python tournament. These distributions are highly skewed, indicating that the tournaments do not sample interactions uniformly across behavioural space.

This naturally raises further questions. Should an "objective" tournament sample interactions uniformly across behavioural space? Should it instead balance different levels of cooperation, reciprocity, exploitability, or payoff asymmetry? More generally, what properties should define a representative interaction environment for evaluating repeated game strategies?

We believe this is an important open question.

Changes:

We have added a discussion of this point in the manuscript. We also include new figures in the Supplementary Information (S7) illustrating the distributions of interaction cooperation rates and payoffs in both the original and expanded tournaments, highlighting their highly skewed structure.

> Overall, this manuscript successfully revisits and extends a foundational work in evolutionary game theory and serves as a valuable case study in computational reproducibility. I recommend the manuscript for publication after minor revisions.

Reply:

We thank the reviewer.

Reviewer 2

> The authors reproduced one of the famous tournaments conducted by Robert Axelrod about half a century ago. Axelrod's main findings were reconfirmed. The authors also tested some variations, e.g., by using an extended set of strategies and adding noise. This is an interesting paper. Reproducibility is an important part of academic integrity, and the preservation of this historical event is a meaningful activity. As long as this lies

> within the scope of this journal, I would like to recommend accepting this manuscript for publication in communications ai & computing.

Reply:

We thank the reviewer for their careful and thoughtful reading of our manuscript, as well as for their encouraging remarks about its contributions.

This work falls naturally within social computing and computational social science, as it studies how interaction environments shape cooperation and collective outcomes in repeated social dilemmas. By revisiting and extending Axelrod's tournaments through large-scale computational experiments, we examine how strategic behaviour, reciprocity, and opponent composition influence tournament outcomes and the emergence of cooperation.

We believe Communications AI & Computing is an excellent fit for this work because the manuscript combines large-scale computational modelling, reproducible software infrastructure, historical computational reproducibility, and the analysis of social interaction systems within the context of social computing.

Reviewer 3

> This outstanding paper faithfully revives Axelrod's second IPD tournament by compiling the original 63 Fortran submissions (with only compiler fixes) and integrating them into Axelrod-Python. It reproduces the core result, TFT (k92r) wins, while uncovering a bug in Champion (k61r) and showing k42r wins 31.5% of replications. Extensions to noisy tournaments (1%) and massive 272-strategy tournaments (with Axelrod-Python + Stewart & Plotkin) reveal TFT's robustness in cooperative settings but its vulnerability in diverse/noisy environments.

Reply:

We thank the reviewer for their detailed and positive assessment of our work. We are grateful for their recognition of the contributions of our paper, as well as for their constructive feedback on how to further strengthen the manuscript.

> None that threaten validity, but two points merit clarification or strengthening for publication: (1) The authors infer the fifth length as 156 from the reported average of 151 moves. This is reasonable but not 100% certain (Axelrod's original announcement may have used slightly different probabilistic sampling). Perhaps a short paragraph acknowledging this (and perhaps providing sensitivity results with alternative length sets, e.g., 63, 77, 151, 200, 308 or full probabilistic sampling) would be useful.

Reply:

Our choice of a match length of 156 follows directly from Axelrod's reported average match length of 151 moves under the five tournament lengths used in the second tournament. However, the incomplete description of the match lengths in the original paper introduces some uncertainty regarding the precise implementation of the tournament.

To address this point, we now additionally perform a sensitivity analysis in which matches end probabilistically rather than after a fixed number of turns. Specifically, we consider a continuation probability corresponding to an average match length of 151 turns.

Overall, we find that the main conclusions remain unchanged under probabilistic ending conditions. In particular, the ranking of strategies remains consistent with the original analysis, and *TFT* continues to perform robustly.

Changes:

...

> (2) The 78+ million tournaments are subsampled from the full interaction matrix. Please clarify the subsampling method (random draws of the added strategies?) and report confidence intervals or standard errors on the proportions (especially the non-monotonic TFT behaviour).

Reply:

We have substantially revised the extra invitations section in response to Reviewer 1's comments, which we believe now makes the construction of these tournaments much clearer.

For each subset size, we consider all possible combinations of strategies drawn from the given strategy set. Thus, the tournaments are not generated through random subsampling of additional strategies; rather, we exhaustively enumerate all possible subsets up to the specified subset size. As a result, the reported winning probabilities correspond to deterministic frequencies over the full combinatorial space of tournaments considered, rather than estimates obtained from a random sample.

We agree, however, that the original text did not explain this procedure clearly enough. We have therefore clarified the construction of the tournaments and expanded the discussion of the observed non-monotonic behavior of *TFT* in the revised manuscript.

Changes:

We have clarified the construction of the extra invitation tournaments in the manuscript.

> Minor issues:
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> (1) "k42r ranked only 3th" → "3rd".
>
> (2) "78,760,605 tournaments", please double-check arithmetic.

Reply:

We have corrected the typo ("3th" to "3rd"). We also clarified that the reported number of tournaments refers to the sum of all combinations obtained by adding between one and four strategies. The correct total is 78,760,569 tournaments.

Changes:

Corrected in the revised manuscript.

> This is a high-quality, timely, and impactful contribution. I strongly recommend acceptance after these light revisions. The paper will be widely read and cited, both for its scientific findings and as a case study in digital preservation of historical computational science.

Reply:

We thank the reviewer.